

EE/CS 172/130
Digital Logic and Design

Topic 6 – Karnaugh Map

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Karnaugh Map

- A Karnaugh map (K-map) is a systematic arrangement of cells in the form of a chart or grid, where each cell corresponds to one of the 2^n possible minterms or maxterms of a Boolean expression with n variables.
- It serves as a graphical technique for the simplification of Boolean expressions, reducing them to their minimal form.
- Although there are multiple approaches to constructing a K-map, we will adopt a single, standardized method to ensure clarity and efficiency in the simplification process.
- In this course, we will use minterm maps and follow this important property:
when moving from one cell to the next along any row or column, exactly one variable in the product term changes its value.

Constructing a K-map

Step 1: Form the Map

2 input variables

- Number of cells = $2^2 = 4$

		B	
		$\bar{B}$	B
A	$\bar{A}$	0 $\bar{A}\bar{B}$	1 $\bar{A}B$
	A	2 $A\bar{B}$	3 AB

3 input variables

- Number of cells = $2^3 = 8$

		BC			
		$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
A	$\bar{A}$	0 $\bar{A}\bar{B}\bar{C}$	1 $\bar{A}\bar{B}C$	3 $\bar{A}BC$	2 $\bar{A}B\bar{C}$
	A	4 $A\bar{B}\bar{C}$	5 $A\bar{B}C$	7 ABC	6 $AB\bar{C}$

Constructing a K-map

Step 1: Form the Map

4 input variables

- Number of cells = $2^4 = 16$

<i>AB</i>	<i>CD</i>			
	<i>$\bar{C}\bar{D}$</i>	<i>$\bar{C}D$</i>	<i>CD</i>	<i>$C\bar{D}$</i>
<i>$\bar{A}\bar{B}$</i>	0 <i>$\bar{A}\bar{B}\bar{C}\bar{D}$</i>	1 <i>$\bar{A}\bar{B}\bar{C}D$</i>	3 <i>$\bar{A}\bar{B}CD$</i>	2 <i>$\bar{A}\bar{B}C\bar{D}$</i>
<i>$\bar{A}B$</i>	4 <i>$\bar{A}B\bar{C}\bar{D}$</i>	5 <i>$\bar{A}B\bar{C}D$</i>	7 <i>$\bar{A}BCD$</i>	6 <i>$\bar{A}BC\bar{D}$</i>
<i>AB</i>	12 <i>$AB\bar{C}\bar{D}$</i>	13 <i>$AB\bar{C}D$</i>	15 <i>$ABCD$</i>	14 <i>$ABC\bar{D}$</i>
<i>$A\bar{B}$</i>	8 <i>$A\bar{B}\bar{C}\bar{D}$</i>	9 <i>$A\bar{B}\bar{C}D$</i>	11 <i>$A\bar{B}CD$</i>	10 <i>$A\bar{B}C\bar{D}$</i>

Constructing a K-map

Step 2: Place outputs of the Boolean expressions in the cell input variables

▪ $F(A, B, C) = \Sigma(0, 2, 3, 7)$

$A \backslash BC$				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0 1	1 0	3 1	2 1
A	4 0	5 0	7 1	6 0

▪ $F(A, B, C) = \pi(0, 1, 5, 6)$

$A \backslash BC$				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0 0	1 0	3 1	2 1
A	4 1	5 0	7 1	6 0

▪ $F(A, B, C) = A + \bar{B}\bar{C}$
 $= \Sigma(0, 4, 5, 6, 7)$

$A \backslash BC$				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0 1	1 0	3 0	2 0
A	4 1	5 1	7 1	6 1

Grouping in K-maps

Grouping rules

- Encircle groups of adjacent 1s.
- The number of 1s in each group must be a power of 2 (1, 2, 4, 8, ...).
- Always form the largest possible groups.
- Rightmost and leftmost cells are considered adjacent.
- Topmost and bottommost cells are considered adjacent.
- Use the minimum number of groups required to cover all 1s.
- Groups may overlap, if necessary.
- Each group must include at least one unique 1 that is not covered by any other group.

Reading the simplified expression

1. For each group, identify the variables that remain unchanged within that group.
2. Write the corresponding product term, and OR all product terms to obtain the simplified expression.

Practice Problems

- $F(A, B, C) = \sum(0, 2, 3, 7)$
 $= BC + \bar{A}\bar{C}$
- $F(A, B, C) = \sum(0, 2, 4, 5, 6, 7)$
 $= A + \bar{C}$
- $F(A, B, C) = \pi(0, 1, 5, 6)$
 $= \bar{A}B + BC + AB\bar{C}$
- $F(A, B, C) = AC + \bar{B}\bar{C} + ABC$
 $= A + \bar{B}\bar{C}$
- $F(A, B, C, D) = \sum(3, 7, 11, 15)$
 $= CD$
- $F(W, X, Y, Z) = \sum(0, 2, 4, 7)$
 $= \bar{W}\bar{Y}\bar{Z} + \bar{W}\bar{X}\bar{Z} + \bar{W}XYZ$
- $F(A, B, C, D) = \sum(1, 3, 5, 7, 11, 15)$
 $= CD + \bar{A}D$
- $F(A, B, C, D) = \sum(1, 3, 5, 7, 10, 11, 14, 15)$
 $= \bar{A}D + AC$
- $F(A, B, C, D) = \pi(12, 13, 15)$
 $= \bar{A} + C\bar{D} + \bar{B}$
- $F(A, B, C, D) = \sum(8, 11, 12, 14, 15)$
 $= A\bar{C}\bar{D} + ACD + ABC/AB\bar{D}$
- $F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 9, 10, 11)$
 $= \bar{B}\bar{D} + A\bar{B} + \bar{A}BD$

Practice Problems

- $F(A, B, C, D) = \Sigma(0, 2, 4, 5, 7, 10)$
 $= \bar{A}\bar{C}\bar{D} + \bar{A}BD + \bar{B}C\bar{D}$
- $F(A, B, C, D) = \Sigma(0, 1, 2, 3, 6, 7, 12, 13, 14, 15)$
 $= \bar{A}\bar{B} + AB + BC/\bar{A}C$
- $F(A, B, C, D) = (\bar{A} + \bar{B} + C + D)(A + \bar{B} + C + D)(A + B + C + \bar{D})(A + B + \bar{C} + \bar{D})$
 $(\bar{A} + B + C + \bar{D})(A + B + \bar{C} + D)$
 $= \pi(1, 2, 3, 4, 9, 12)$
 $= \overline{B\bar{C}\bar{D} + \bar{B}\bar{C}D + \bar{A}\bar{B}C}$
 $= (\bar{B} + C + D)(B + C + \bar{D})(A + B + \bar{C})$
 $= AC + BC + BD + \bar{B}\bar{C}\bar{D}$

Don't Care Conditions

- In many practical logic designs, there are certain combinations of input variables that will never occur. These impossible input combinations are marked as **d** or **X** in truth tables and K-maps.
- In a K-map, these don't care conditions can be:
 - Treated as 1 to form a larger group and achieve a simpler expression, or
 - Treated as 0 and ignored entirely.
- Since it does not affect the final output whether they are considered 1 or 0, they are called don't care conditions.

Practice Problems

- $F(A, B, C, D) = \sum(1,3,7,11,15) + d(0,2,5)$
 $= \bar{A}\bar{B} + CD$
- $F(A, B, C, D) = \sum(0,2,4,6,8) + d(10,11,12,13,14,15)$
 $= \bar{D}$
- $F(A, B, C, D) = \sum(0,1,2,9,11) + d(8,10,14,15)$
 $= A\bar{B} + \bar{B}\bar{C} + \bar{B}\bar{D}$
- $F(A, B, C, D) = \sum(0,6,8,13,14) + d(2,4,10)$
 $= C\bar{D} + \bar{B}\bar{D} + AB\bar{C}D$
- $F(A, B, C, D) = \sum(0,1,4,5,6) + d(2,3,7)$
- $F(A, B, C, D) = \sum(5,6,7,12,14,15) + d(3,9,11,15)$
- $F(A, B, C, D) = \sum(2,4,7,10,12) + d(0,6,8)$

Practice Problems

- Design a logic circuit that outputs 1 whenever the BCD input $ABCD$ represents an odd digit.
 $F(A, B, C, D) = \sum(1,3,5,7,9) + d(10,11,12,13,14,15) = D$
- Design a logic circuit that detects invalid BCD codes (10–15) and outputs 1 for these cases. Let the input be $ABCD$.

$$F(A, B, C, D) = \sum(10,11,12,13,14,15) = AB + AC$$

5 Variable K-map

Construction

<i>AB</i> \ <i>CDE</i>									
		$\bar{C}\bar{D}\bar{E}$	$\bar{C}\bar{D}E$	$\bar{C}D\bar{E}$	$\bar{C}DE$	$CD\bar{E}$	CDE	$C\bar{D}E$	$C\bar{D}\bar{E}$
$\bar{A}\bar{B}$		0 $\bar{A}\bar{B}\bar{C}\bar{D}\bar{E}$	1 $\bar{A}\bar{B}\bar{C}\bar{D}E$	3 $\bar{A}\bar{B}\bar{C}D\bar{E}$	2 $\bar{A}\bar{B}\bar{C}DE$	6 $\bar{A}\bar{B}C\bar{D}\bar{E}$	7 $\bar{A}\bar{B}CDE$	5 $\bar{A}\bar{B}C\bar{D}E$	4 $\bar{A}\bar{B}C\bar{D}\bar{E}$
$\bar{A}B$		8 $\bar{A}B\bar{C}\bar{D}\bar{E}$	9 $\bar{A}B\bar{C}\bar{D}E$	10 $\bar{A}B\bar{C}D\bar{E}$	11 $\bar{A}B\bar{C}DE$	14 $\bar{A}BC\bar{D}\bar{E}$	15 $\bar{A}BCDE$	13 $\bar{A}BC\bar{D}E$	12 $\bar{A}BC\bar{D}\bar{E}$
AB		24 $AB\bar{C}\bar{D}\bar{E}$	25 $AB\bar{C}\bar{D}E$	27 $AB\bar{C}D\bar{E}$	26 $AB\bar{C}DE$	30 $ABC\bar{D}\bar{E}$	31 $ABCDE$	29 $ABC\bar{D}E$	28 $ABC\bar{D}\bar{E}$
$A\bar{B}$		16 $A\bar{B}\bar{C}\bar{D}\bar{E}$	17 $A\bar{B}\bar{C}\bar{D}E$	19 $A\bar{B}\bar{C}D\bar{E}$	18 $A\bar{B}\bar{C}DE$	22 $A\bar{B}C\bar{D}\bar{E}$	23 $A\bar{B}CDE$	21 $A\bar{B}C\bar{D}E$	20 $A\bar{B}C\bar{D}\bar{E}$

5 Variable K-map

Mapping

- Represent the 5-variable K-map as two 16-cell halves: one half for the third variable = 0, and the other half for the third variable = 1.
- Inside each half, form groups of 1s as per the rules.
- After grouping within halves, overlay (fold) the two halves so corresponding cells that differ only in the third variable can be combined into larger groups.
- Combine any groups that completely overlap or that form valid larger groups across the folded halves, still respecting power-of-two sizes. Groups may overlap and may wrap around edges.
- For every final group, write the product term formed by the variables that do not change in that group; OR all product terms to obtain the simplified Boolean expression.

Practice Problems

- $F(A, B, C, D, E) = \sum(0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31)$
 $= \bar{A}\bar{B}\bar{E} + BE + A\bar{D}E$
- $F(A, B, C, D, E) = \sum(0, 1, 2, 3, 4, 5, 6, 7, 10, 19, 23, 25, 27, 29, 31)$
 $= \bar{A}\bar{B} + ABE + \bar{A}\bar{C}D\bar{E} + ADE/\bar{B}DE$
- $F(A, B, C, D, E) = \sum(15, 23, 27, 29, 30, 31)$
 $= BCDE + ACDE + ABCD + ABCE + ABDE$
- $F(A, B, C, D, E) = \sum(1, 5, 6, 7, 14, 15, 16, 20, 22, 23, 30, 31)$
 $= CD + \bar{A}\bar{B}\bar{D}E + A\bar{B}\bar{D}\bar{E}$
- $F(A, B, C, D, E) = \sum(0, 1, 4, 8, 12, 13, 15, 16, 17, 23, 29, 31)$
 $= \bar{A}\bar{D}\bar{E} + BCE + \bar{B}\bar{C}\bar{D} + ACDE$
- $F(A, B, C, D, E) = \sum(0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30)$
 $= \bar{E}$